

Where Nobody Lives

Building a national map of the census blocks that hold no people

Democracy's Data

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A census block is the smallest piece the Census Bureau cuts the country into, and the reason it exists is arithmetic. Every other geography the Bureau publishes — a tract, a city, a school district, a congressional district — has to be expressible as some pile of blocks, because counts add upward but cannot be divided downward. Make the smallest piece too big and whole classes of question stop having answers. The block is the Bureau's answer to how small the smallest piece has to be.

The Bureau draws them, not the states, and it draws them along things a person could stand next to: streets, streams, railroad tracks, shorelines, and the legal boundaries of counties, cities and reservations.¹ States do get a say. Through the Redistricting Data Program they can suggest where block boundaries should fall, so that blocks nest inside the precincts they actually run elections in. But the pen is federal, and a block is drawn for the convenience of counting rather than to mark anything a resident would recognize.

There are 8,174,955 of them and they are small. The median block with any land on it covers 455,841 square feet — about 10.5 acres — and holds 14 people.

The extremes run a long way from that median, in both directions and not together. The largest block in the country by area covers 9,226 square miles in North Slope Borough, Alaska, more land than 10 states and state equivalents including Vermont, whose 9,217 square miles it exceeds. Its population is 0: the biggest block in the country is an empty one. The largest by population is 8,727 people in Los Angeles County, California, on 38 acres.

That pairing is this brief's subject. 2,368,443 blocks — 29.0% of every block in the country — recorded nobody at all in 2020. That is not a defect in the count. It is what happens when you cut a country into pieces small enough that most pieces are a lake, an interchange, a runway, a cemetery, or a stretch of national forest.

This brief is a walkthrough. It shows where the data came from, what was done to it, and what the resulting map does and does not say. The point is not the map at the end; it is that you could rebuild it.

¹ The rule and the definition are in the Bureau's glossary entry for census block; the visible-feature principle is why a river or an interstate so often has blocks of its own on either side of it.

01 Where the data came from

One file, and it holds both halves of the problem.

The shapes. Block boundaries come from TIGER/Line, the Bureau's digital outlines of every census unit, in its 2020 release. The block files are very large and are downloaded separately by the build rather than kept beside this chapter.

The count rides along with the shape. Every block record in those files carries a POP20 field: the number of people the 2020 decennial census counted in that block. It is not an

estimate and not a sample. It is the decennial count itself, the same number the Bureau publishes in the P.L. 94-171 Redistricting Data Summary File and delivers to every state legislature for redistricting. The Bureau prepends it to the geography, so a block's boundary and its population arrive in the same file.

That is why this brief is short. There is no second source to fetch, no key to register, and no join to get wrong. The same counts are also served one variable at a time by the Census Data API, which is a chapter of its own; for a map of every block in the country, the file that already carries the number is the shorter road.

Three properties make this the right count. It is complete rather than sampled, so there are no margins of error below and no allowance for how far off an estimate could be, because nothing is estimated. Its unit, the census block, is the smallest the Bureau publishes, so everything larger can be built out of it. And it is the operative number rather than an academic one: it is what district maps are drawn from.

Both roads were walked here before choosing one. Read from POP20 and read from the API, the two agree exactly — the same 8,174,955 blocks, the same 2,368,443 empty ones, the same 334,735,155 people, and the same count at every one of the 1,001 rungs of the distribution below.

02 How the map was made

The pipeline is four steps, and each one throws something away on purpose.

1. **Read** each state's block file and keep the blocks whose POP20 is zero. One state at a time, so that a single state is in memory rather than the whole country.
2. **Drop the water.** A block that is all water reports zero people truthfully, but drawing those fills the Great Lakes and every coastal bay with solid color. Blocks with no land area are set aside here, and counted separately.
3. **Dissolve** the surviving blocks into one shape. Touching empty blocks become one region, which is honest here: the boundary between two adjacent empty blocks is not a feature of the world, it is an artifact of where the Bureau put its lines.
4. **Simplify** with mapshaper, project to Albers USA, and round the coordinates to whole units of the figure's own grid.

Simplification is the step that deserves suspicion, so here are its terms. The national layer at full resolution is about 200 MB, far too large to put in a web page. What ships is 2% of the original vertices: 321,181 points in 18,200 polygons, about 1 MB. At the width this figure is drawn, one unit of the grid is roughly 2.3 kilometers, so detail finer than that could not be seen even if it were kept. What is lost is the exact shape of any individual block. What survives is where the empty land is, which is the only claim the figure makes.

Two things in steps 3 and 4 are easy to get wrong and were got wrong here first. Dissolving produces polygons with **holes** — a populated town inside an empty county is an interior ring — and a ring drawn on its own is painted as fill rather than subtracted, which turns Nevada into a solid rectangle. Both renderings below therefore draw one path per polygon with all of its rings together, under an even-odd fill rule. And rounding coordinates has to happen *after* projecting, not before: rounding longitude and latitude to whole numbers rounds them to whole degrees, and the country dissolves.

The built tables are small and readable: `state_blocks.csv`, one row per state; `block_pop_hist.csv`, a count of blocks at each population from 0 to 1,000;

zero_land.csv, land area of the empty blocks by state; block_size.csv, how big a block is and the two extremes; facts.csv, the national totals quoted in this text; and the figure geometry in map_zero.csv, map_states.csv and map_frame.csv, stored as integer figure coordinates so the web version and the print version draw the same shapes from the same numbers.

Level	Where	Blocks	Blocks with nobody	Percent
Nation	50 states, DC, Puerto Rico	8,174,955	2,368,443	29.0%
Most empty	Alaska	28,568	16,803	58.8%
Least empty	Puerto Rico	41,987	5,417	12.9%

03 What the map shows

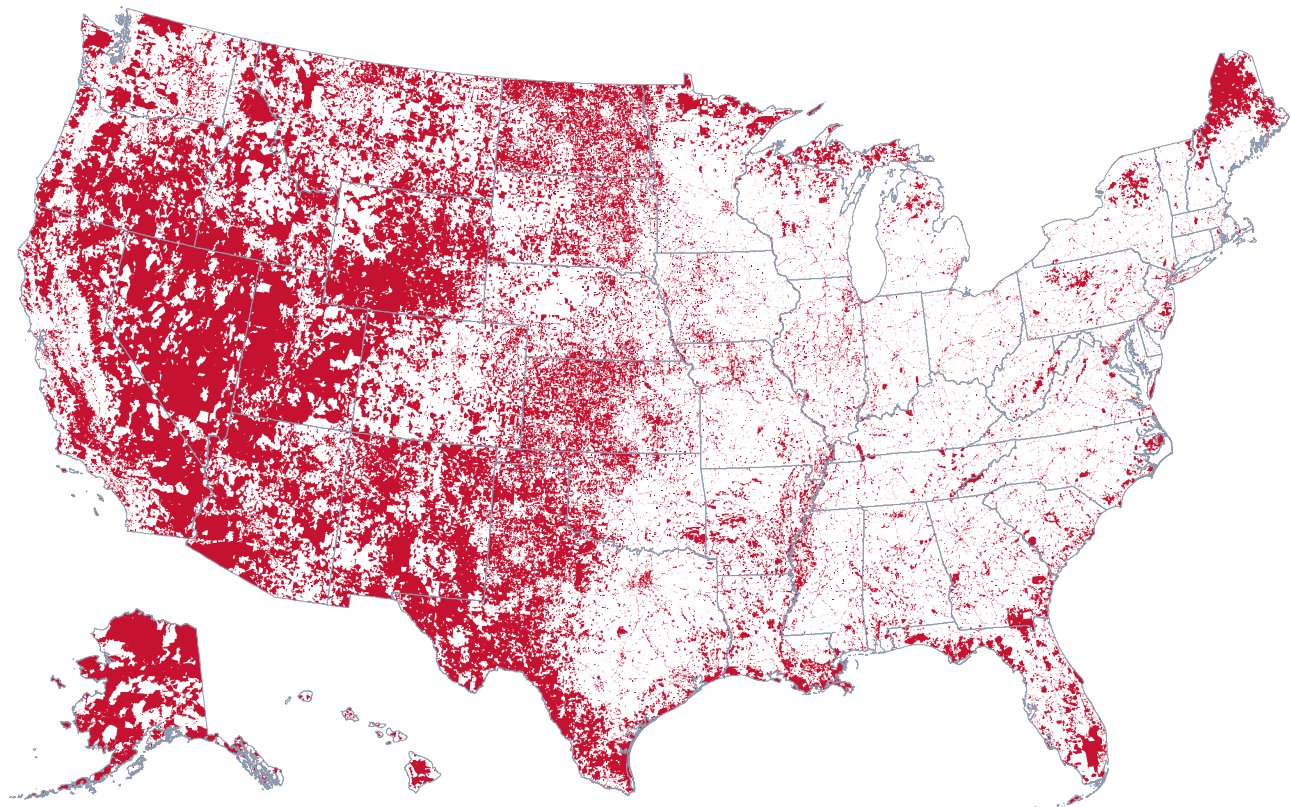


Figure 1. Every census block in the United States that recorded no residents in 2020.

Red is empty, white is inhabited land, and the map draws the 2,130,094 empty blocks that have land area. A single flat color fits the question, because the claim is about where, not how much. In the web edition, hovering a state gives its population, its block count, the share of those blocks that are empty, and how much of its land is unpopulated.

Read west to east and the map turns into a history of land ownership. The solid masses through Nevada, Idaho and eastern Oregon are federal land: national forest, Bureau of Land Management range, military reserve. The sharp vertical seam around the hundredth meridian is where homestead-scale agriculture stopped being viable. In the East the pattern inverts into a fine red lacework that traces rivers, interstates and the edges of reser-

voirs, because east of the Mississippi the empty blocks are mostly small and linear rather than large and areal.

The number under the map is the one that surprises people. Those 29.0% of blocks cover about 1,465,387 square miles, roughly 41.5% of the land area of the United States. **Blocks are not land.** A statistic about how many blocks are empty and a statistic about how much of the country is empty are different statistics, and the gap between 29.0% and 41.5% is the size of the difference.

04 How big is a census block, anyway?

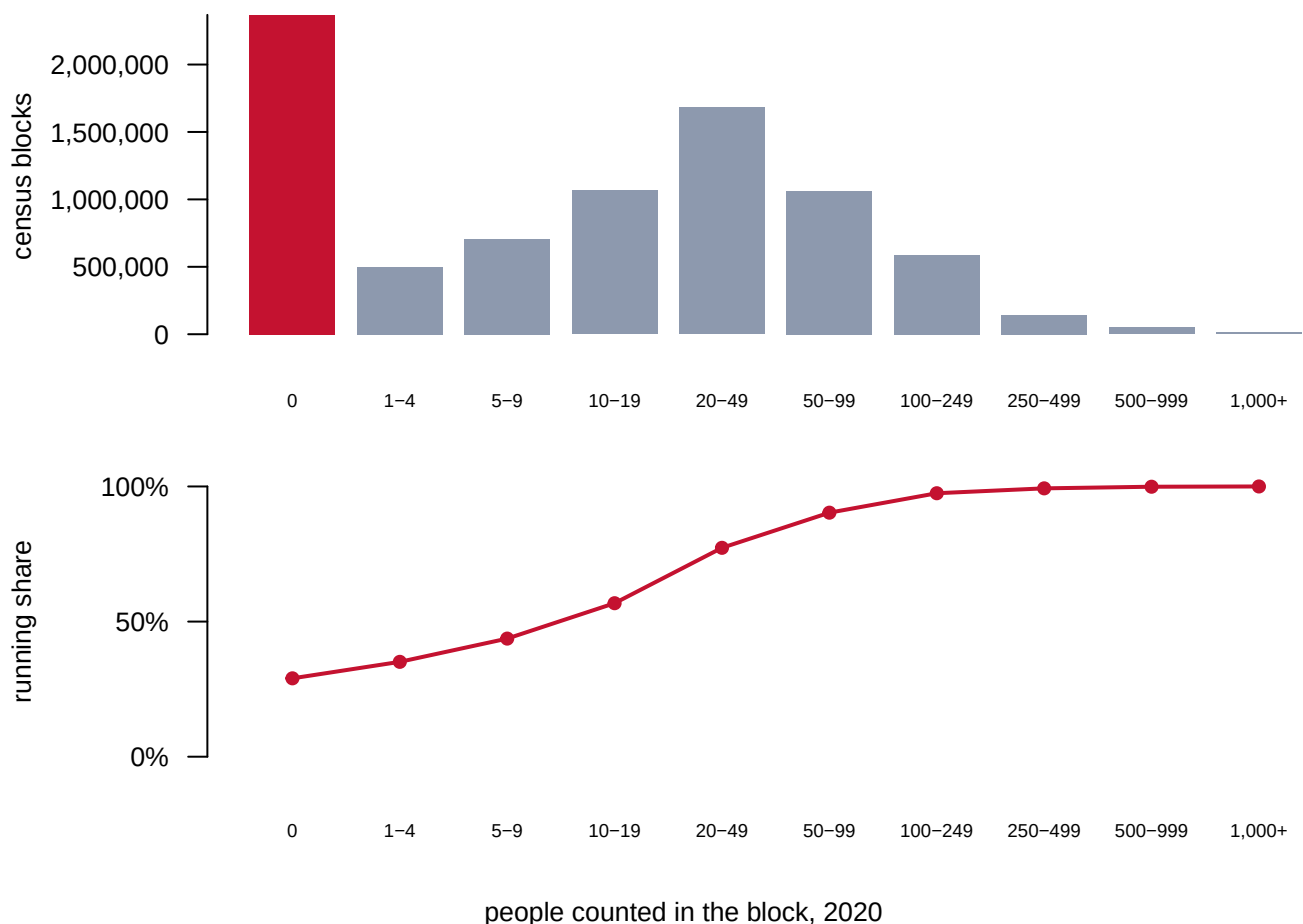


Figure 2. How many people live in a census block. Blocks grouped by population, from empty to a thousand and over; the red bar is the 2,368,443 with nobody. Bins widen toward the tail because block populations do: binned this way the whole range fits one axis that starts at zero, so every bar's length is its count and nothing has to be logged or broken. The panel beneath is the running share — the portion of all blocks at or below each bin. Hover a bin in the web edition for its exact count.

The shape says the block is a small thing. The median block in the country holds 14 people. Among blocks with anybody at all, the median is 28. 45.3% of all blocks hold ten people or fewer, and the running share passes half by the 20-49 bin. A census block is not a neighborhood; it is closer to a city block, which is what it was named for.

This matters for reading any block-level statistic. A percentage computed on a unit holding fourteen people is mostly noise, and the Bureau's disclosure-avoidance noise is deliberately largest, in relative terms, exactly here.

05 Which states are emptiest

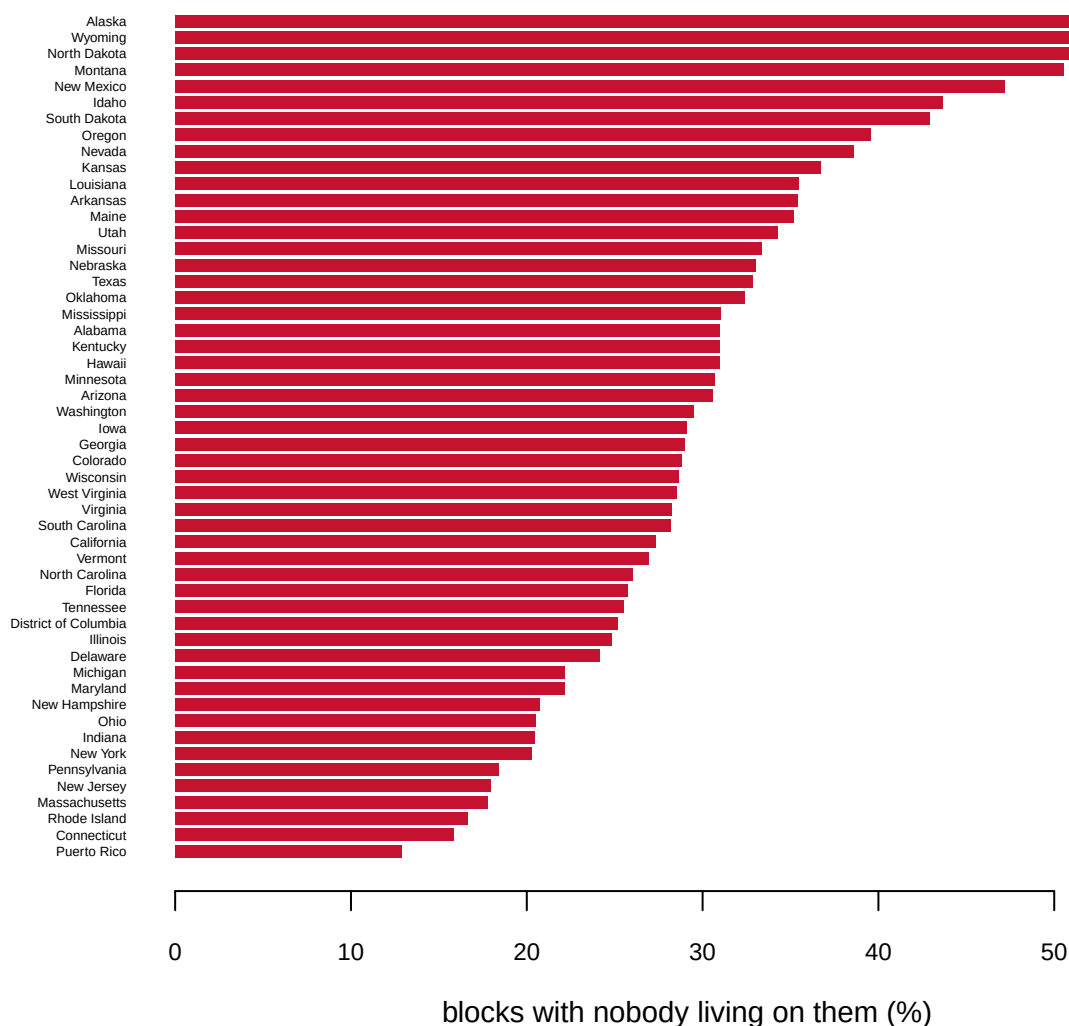


Figure 3. The share of each state's blocks that recorded nobody. Sorted rather than alphabetical, because the ordering is the finding. Hovering a bar in the web edition gives the counts underneath the percentage.

The range runs from Alaska at 58.8% to Puerto Rico at 12.9%. It is tempting to read this as a map of emptiness, and it is partly that. But it is also a map of how the Bureau cut each state. Blocks are drawn along visible features, so a state with a dense grid of section-line roads gets many small blocks whether or not anyone lives on them, and a state with a lot of water gets a lot of water blocks. Alaska is genuinely empty; it also has an enormous coastline and river network, each stretch of which is its own block.

06 Is this just a map of density?

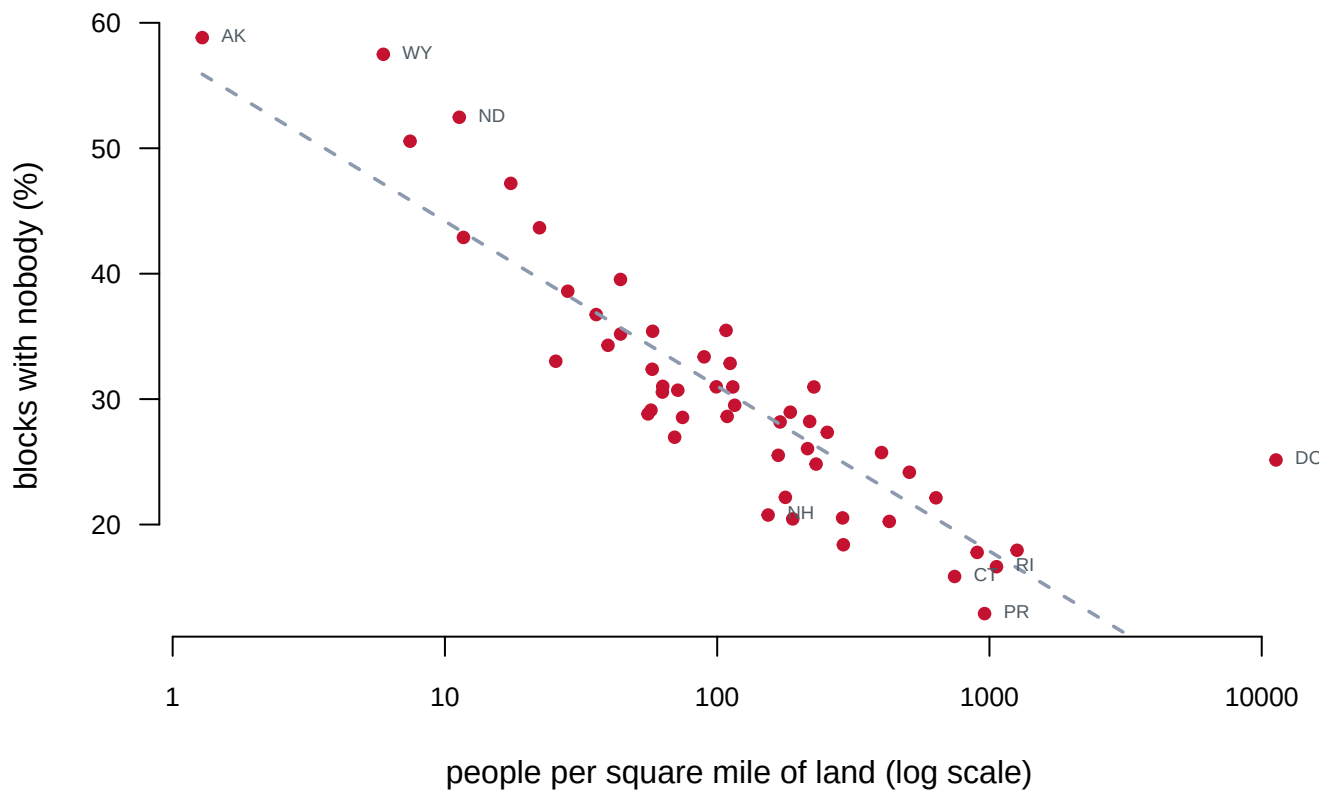


Figure 4. Each state's emptiness against its population density. One dot per state, density on a log scale because state densities span four orders of magnitude, from 1.3 people per square mile in Alaska to 11,286 in the District of Columbia. The dashed line is the least-squares fit. Only the extremes and the states furthest from the line are labelled; hover any dot in the web edition for the rest.

The honest answer is: largely, but not entirely, and the gap is where the interest is. The correlation between log density and the share of a state's blocks that are empty is -0.87 , so density accounts for about 76% of the variation between states. That is a strong relationship and it is the one you would guess.

Two things complicate it. The first is the log scale, which is doing real work: on untransformed density the same correlation is only -0.20 . Emptiness tracks the *order of magnitude* of a state's density, not its level, so the distance from Wyoming to Ohio matters far more than the distance from Ohio to New Jersey.

The second is the District of Columbia, the point stranded at the right. It is the densest place in the country at 11,286 people per square mile, and 25.1% of its blocks are still empty – the Mall, the rail yards, Rock Creek Park, the river. Drop it and the correlation tightens to -0.93 . It is one observation, and it is the whole argument in one dot: **a place can be as dense as anywhere in America and still be full of empty blocks**, because emptiness is a property of how the land was cut, not only of how many people live on it.

07 What this brief cannot tell you

The 29.0% is a fact about blocks, not about people or land, and the three do not track each other. Nothing here says how much of the country is *uninhabitable*, only which units recorded no usual residents on 1 April 2020. A block containing a large prison, a dormitory closed for the pandemic, or a seasonal resort counted empty that day may be busy most of the year.

The map is drawn from the land blocks. 238,349 of the empty blocks are water and are not drawn, because including them fills the Great Lakes and every coastal bay with solid color and makes the map a map of hydrography. That is a presentation choice, and it is reversible; the land-area figures are unaffected either way, because the Bureau's ALAND20 already excludes water.

Empty is not the same as thinly settled, and this is not a population density map, though Figure 4 shows how much of one it is. Density is people divided by area, a continuous quantity every block has some value of. Emptiness is a threshold applied to a unit: zero, or not zero. A solid red county on this map means its blocks crossed that threshold, not that nobody lives nearby.

Simplification means no individual block on this map is the right shape. Do not measure anything off it. The built tables carry the counts; the figure carries the pattern.

And zero is a published number, not a raw one. The 2020 counts passed through disclosure avoidance before release, which can move a small count by a little. At the block level that noise is proportionally largest, though a block that is a reservoir was not going to be populated either way.

08 What you should have learned

- The census block is small: the median holds 14 people, and 45.3% hold ten or fewer. Percentages computed on units that size are fragile.
- 29.0% of blocks are empty but they cover about 41.5% of the land. Counting units and measuring area are different operations, and “most of the country is empty” and “most blocks are empty” are different claims.
- A join that matches 100% is a test you passed, not a formality. Identifiers are text.
- Every step that made this map — filtering, dissolving, simplifying — discarded something. A figure is trustworthy when you can say what it discarded and why, not when nothing was discarded.

09 Extensions

- `state_blocks.csv` — divide pop by blocks for each state. Which states put the fewest people in a block, and is that the same list as the emptiest states in Figure 3?
- `state_blocks.csv` joined to `zero_land.csv` — rank states by `pct_zero`, then by `zero_land_sqmi`. Name a state that is high on one list and low on the other, and say what the Bureau did to produce that gap.
- `block_pop_hist.csv` — the last row is a bucket, every block of 1,000 people or more. How many blocks are in it, and what kind of place holds that many people inside one block?

- `block_pop_hist.csv` — compute the share of the national population living in blocks of ten people or fewer. Compare it with the 45.3% of *blocks* that size. Explain the difference to someone in one sentence.
- **Stretch.** Download one state's block file and check its POP20 against the same counts served by the API. Report how many disagree, and decide which you would trust in print.
- **Stretch.** Rebuild Figure 1 for one state without the dissolve step, drawing each empty block separately. Say what the dissolved version hides and what it clarifies.

10 Sources

Data

- U.S. Census Bureau, 2020 Census Redistricting Data (P.L. 94-171) Summary File, served through the Census Data API as the 2020/dec/p1 dataset; variable P1_001N is total population, and summary level block is the Bureau's smallest published unit. <https://api.census.gov/data/2020/dec/p1> and <https://www.census.gov/data/developers/data-sets/decennial-census.html>
- U.S. Census Bureau, TIGER/Line Shapefiles 2020, TABBLOCK20 layer — the boundary of every 2020 tabulation block, one file per state, carrying GEOID20, ALAND20 and AWATER20. <https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/>
- U.S. Census Bureau, Cartographic Boundary Files 2020, state layer at 1:500,000 — generalized outlines drawn for mapping rather than measurement, clipped to the shoreline, used here only as the frame of Figure 1. <https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`block_pop_hist.csv`, `block_size.csv`, `facts.csv`, `map_frame.csv`, `map_states.csv`, `map_zero.csv`, `state_blocks.csv`, `zero_land.csv`

Method

- Bloch, M. *mapshaper* — the simplification, dissolve and projection steps in Figure 1 are all mapshaper commands; its Visvalingam implementation is what “2% of the vertices” means here. <https://github.com/mbloch/mapshaper>
- Bostock, M. *D3* — the three figures in the HTML edition are plain SVG built with D3 scales and axes. <https://d3js.org/>

Documentation

- U.S. Census Bureau, “Glossary: Census Block” — the definition, and the rule that blocks are bounded by visible features and by legal boundaries. <https://www.census.gov/programs-surveys/geography/about/glossary.html>
- U.S. Census Bureau, “Disclosure Avoidance and the 2020 Census” — why a published block count is not a raw block count. <https://www.census.gov/programs-surveys/decennial-census/decade/2020/planning-management/process/disclosure-avoidance.html>

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any

starting files. Please do the following and show your work.

THE FIRST SOURCE. The U.S. Census Bureau's Data API, 2020 Decennial Census

P.L. 94-171 Redistricting Summary File:

<https://api.census.gov/data/2020/dec/pl>

Free, but it needs a key from

https://api.census.gov/data/key_signup.html.

The variable for total population is P1_001N. The API will not return every

block in a state at once: a block query has to name a state AND a county, so

covering the country takes about 3,100 requests. Block GEOIDs are 15-character strings – read them as text, never as numbers, or a 15-digit

identifier becomes 1.3121e+14 and leading zeros vanish.

THE SECOND SOURCE. TIGER/Line 2020 tabulation block shapes, one zipped shapefile per state:

https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/tl_2020_<FIPS>_tabblock20.zip

About 10 GB for the whole country. The attribute table carries GEOID20, ALAND20 (land area, square meters) and AWATER20.

WHAT TO BUILD.

1. A table of every census block in the 50 states, DC and Puerto Rico with

its 2020 population. Report the national total, and how many blocks recorded zero.

2. A count of blocks at each exact population from 0 to 1,000, with everything

above 1,000 in one bucket. Report the median block population, and the median among blocks with at least one resident.

3. A per-state table: blocks, blocks with nobody, and the percentage.

4. A national map of the zero-population blocks. Join the zero GEOIDs to the

TIGER shapes state by state, keep the matches, and dissolve touching

blocks together. Report what fraction of the zero blocks have land area

(the rest are water) and how many square miles the land ones cover.

THINGS THAT WILL BITE YOU.

- Check the join. Every GEOID from the API should match a TIGER shape. If it

does not, you read an identifier as a number somewhere.

- Water blocks have zero population and are real. Decide whether they belong

on your map and say which you chose; drawing them fills the Great Lakes.

- Dissolving makes polygons with holes. Draw each polygon's rings together

under an even-odd fill rule, or the holes get painted as fill.

- Round coordinates only AFTER projecting. Rounding longitude and latitude

rounds to whole degrees.

CHECK YOUR WORK. The copies this book used were downloaded in September 2026. If your rebuild matches them, you should find:

- 8,174,955 blocks holding 334,735,155 people, across 52 states and state-equivalents

- 2,368,443 blocks with nobody living on them – 29.0%

- a median block population of 14, or 28 among blocks with at least one resident; 45.3% of blocks hold ten people or fewer

- emptiest Alaska at 58.82%, least empty Puerto Rico at 12.90%

- 2,130,094 of the empty blocks have land area, covering 1,465,387 square miles – about 41% of U.S. land, against 29% of blocks

- every one of the 2,368,443 API identifiers matches a TIGER shape; if yours does not, you read a GEOID as a number

Both sources are finished 2020 census products; they will not change until the 2030 cycle publishes successors.